

9(a)	<i>any six from:</i> 1 (short-term) store of energy ; 2 (energy derived) from, food / respiration / photophosphorylation / chemiosmosis ; 3 transfers energy (to) / (immediate) energy donor (to) ; 4 in <u>all</u>, cells / organisms ; 5 hydrolysis / phosphate loss, releases, energy / 30.5 kJ (per mole) ; 6 reversible reaction / (AMP and) ADP recycled to ATP ; 7 small (molecule) / water soluble, so can move within, cell / cytoplasm ; 8 links energy yielding and energy requiring reactions ; 9 high turnover (rate) ; <i>and any two uses up to question total of 6 from:</i> 10 active transport / action potential / electrical discharge ; 11 muscle contraction ; 12 anabolic reactions / condensation reactions / transcription / translation / DNA replication / Calvin cycle / phosphorylation reactions ; 13 exocytosis / endocytosis / intracellular transport ; 14 bioluminescence ;	6
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Question	Answer	Marks
9(b)	<i>any nine from:</i> <i>anaerobic respiration in, mammalian cells / lactic fermentation (M):</i> 1 pyruvate, is reduced / accepts H / accepts H^+ + electron ; 2 (process uses) reduced NAD ; 3 from glycolysis ; 4 converted to, lactate / lactic acid ; 5 <i>ref. to lactate dehydrogenase</i> ; 6 regeneration of NAD allows glycolysis to continue ; 7 small energy yield / 2 ATP, (from glycolysis) ; <i>difference to, yeast cells / alcoholic fermentation (Y): accept ora mps 8–11</i> 8 M no, decarboxylation / CO_2 removed (to form ethanal) ; 9 M single / one, step / enzyme or Y two, steps / enzymes ; 10 M pyruvate is H acceptor and in Y ethanal is H acceptor ; 11 M reversible / lactate is (later) converted back to pyruvate or Y irreversible or ethanal / ethanol, cannot be converted back to pyruvate ; 12 M needs oxygen / by oxidation / <i>ref. to oxygen debt</i> ;	9